Mathematical Foundations of Political Methodology Lecture 4

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Derivative: Motivation

- In this lecture, we will cover three broad ideas
  - Intuition for the derivative
  - Important tricks for taking derivatives
  - Mean Value Theorem

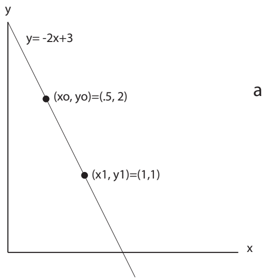
Line and Slope

A line can be represented by the equation

$$y = a * x + b$$

a is the slope and b the y-intercept

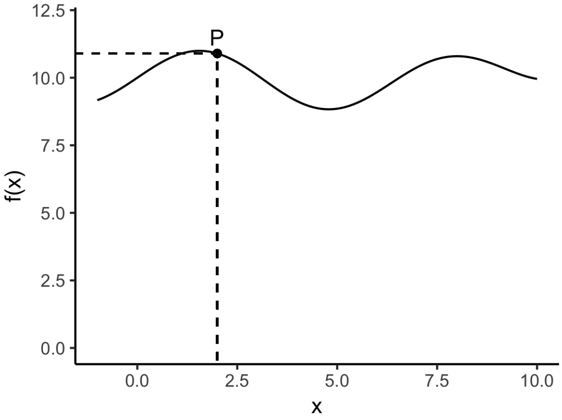
Rise over Run: For any points on the line, (x_o, y_o) and (x_1, y_1), the slope $a = \frac{y_1 - y_o}{x_1 - x_o}$



a =(y1-yo)/(x1-xo)
=(1-2)/(1-.5)
=-2

Slope of a Curve

- Much of the time our functions are not straight lines. In these cases, we add the notion of limits to our definition.
- Consider the slope at the point P.



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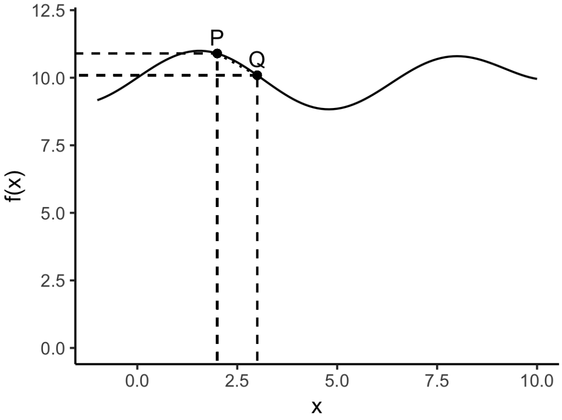


Slope of a Curve

The *slope of the curve* $f(x)$ at point P is the limit of the slope of the line between points Q and P as $Q \to P$

- Consider points $P = (x_0, f(x_0))$ and $Q = (x_1, f(x_1))$.
- Write $x_1 = x_0 + h$ so that we have $x_1 \to x_0$ as $h \to 0$

  $P = (x_0, f(x_0))$ and $Q = (x_0 + h, f(x_0 + h))$.

$$\frac{\text{rise}}{\text{run}} = \frac{y_1 - y_0}{x_1 - x_0} = \frac{f(x_o + h) - f(x_0)}{x_o + h - x_0} = \frac{f(x_o + h) - f(x_0)}{h}$$

As $h \to 0$, $Q \to P$ and we have the *derivative* of $f(x)$ at x_o.

Derivative: Formal Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$.

Derivative Definition

Definition

Let $f : \mathbb{R} \to \mathbb{R}$. *If the limit*

$$\begin{aligned} \lim_{x \to x_0} R(x) &= \frac{f(x) - f(x_0)}{x - x_0} \\ &= f^{'}(x_0) \end{aligned}$$

exists then we say that f *is differentiable at* x_0.
If $f^{'}(x_0)$ *exists for all* $x \in$ *Domain, then we say that* f *is differentiable.*

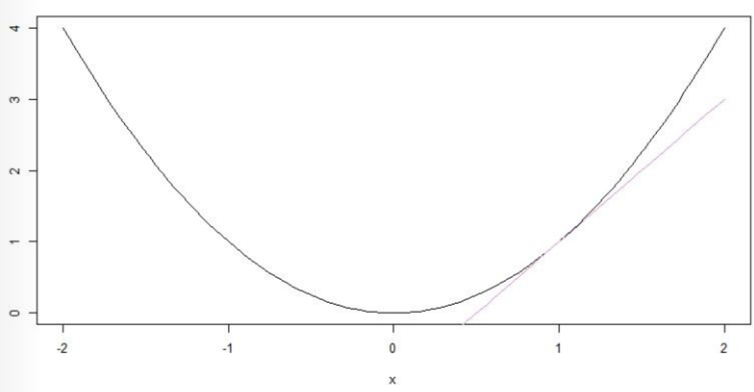
Derivative Examples

- Suppose $f(x) = x^2$ and consider $x_0 = 1$. Then,

$$\begin{aligned}
\lim_{x \to 1} R(x) &= \lim_{x \to 1} \frac{x^2 - 1^2}{x - 1} \\
&= \lim_{x \to 1} \frac{(x-1)(x+1)}{x - 1} \\
&= \lim_{x \to 1} x + 1 \\
&= 2
\end{aligned}$$

The figure looks like:

Derivative Examples



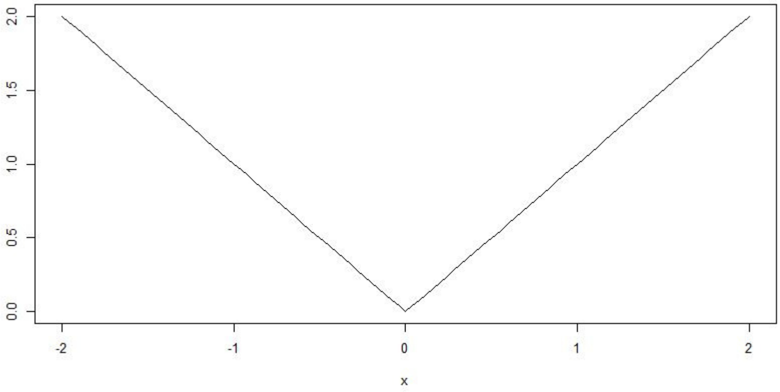
Derivative Existence

- Not all functions have a derivative.
- Suppose $f(x) = |x|$ and consider $x_0 = 0$. Then,

$$\lim_{x \to 0} R(x) = \lim_{x \to 0} \frac{|x|}{x}$$

$\lim_{x \to 0^-} R(x) = -1$, but $\lim_{x \to 0^+} R(x) = 1$.
Given this, we say, $|x|$ not differentiable at 0.

Graph of $|x|$



Derivative and Continuity

$|x|$ is continuous but not differentiable. If a function is differentiable, then it is continuous.

Theorem

Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable at x_0. Then f is continuous at x_0.

Our proof strategy:

- We need $\lim_{x \to x_0} f(x) = f(x_0)$ (recall last lecture).
- Use direct proof: Start with f(x) that is differentiable.
- Manipulate till we get the formula for the rate of change $R(x) = \frac{f(x) - f(x_0)}{(x - x_0)}$.
- Take the limit, use limit properties, get $f(x_0)$.

Proof of Derivative and Continuity

Proof.

Consider some differentiable $f(x)$:

$$
\begin{aligned}
f(x) &= f(x) - f(x_0) + f(x_0) \\
f(x) &= \frac{f(x) - f(x_0)}{(x - x_0)}(x - x_0) + f(x_0) \\
&= R(x)(x - x_0) + f(x_0)
\end{aligned}
$$

Remember: if $f(x)$ is continuous at x_0 then, $\lim_{x \to x_0} f(x) = f(x_0)$.

$$
\begin{aligned}
\lim_{x \to x_0} f(x) &= \lim_{x \to x_0} [R(x)(x - x_0) + f(x_0)] \\
&= \left(\lim_{x \to x_0} R(x)\right)\left(\lim_{x \to x_0} (x - x_0)\right) + \lim_{x \to x_0} f(x_0) \\
&= f'(x_0)0 + f(x_0) \\
&= f(x_0)
\end{aligned}
$$

- Derivative
- Derivative and Continuity
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

Useful Derivative Rules

Suppose a is some constant, $f(x)$ and $g(x)$ are functions

$$
\begin{aligned}
f(x) = x \quad &; \quad f^{'}(x) = 1 \\
f(x) = ax^{k} \quad &; \quad f^{'}(x) = (a)(k)x^{k-1} \\
f(x) = e^{x} \quad &; \quad f^{'}(x) = e^{x} \\
f(x) = \sin(x) \quad &; \quad f^{'}(x) = \cos(x) \\
f(x) = \cos(x) \quad &; \quad f^{'}(x) = -\sin(x) \\
f(x) = log(x) \quad &; \quad f^{'}(x) = \frac{1}{x}
\end{aligned}
$$

Proof the $exp(x) \equiv e^x$ is its own derivative

Recall $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$\begin{aligned}
exp'(x) &= \lim_{h \to 0} \frac{exp(x+h) - exp(x)}{h} \\
&= \lim_{h \to 0} \frac{exp(x)exp(h) - exp(x)}{h} \\
&= exp(x) \lim_{h \to 0} \frac{exp(h) - 1}{h}
\end{aligned}$$

Proof the $exp(x) \equiv e^x$ is its own derivative (cont.)

$$
\begin{aligned}
\frac{exp(h) - 1}{h} &= \frac{\sum_{n=0}^{\infty} \frac{h^n}{n!} - 1}{h} \\
&= \frac{[\frac{h^0}{0!} + \frac{h^1}{1!} + \frac{h^2}{2!} \ldots] - 1}{h} \\
&= \frac{[\frac{1}{1} + \frac{h^1}{1} + \frac{h^2}{2} \ldots] - 1}{h} \\
&= \frac{[h + \frac{h^2}{2} \ldots]}{h} \\
&= 1 + \frac{h}{2} + \frac{h^2}{3} \ldots
\end{aligned}
$$

Proof the $exp(x) \equiv e^x$ is its own derivative (finish)

$$
\begin{aligned}
exp'(x) &= \lim_{h \to 0} \frac{exp(x+h) - exp(x)}{h} \\
&= exp(x) \lim_{h \to 0} \frac{exp(h) - 1}{h} \\
&= exp(x) \lim_{h \to 0} [1 + \frac{h}{2} + \frac{h^2}{3} \ldots] \\
&= exp(x) * 1 \\
exp'(x) &= exp(x) \qquad \square
\end{aligned}
$$

Derivative in Political Economy

- The policy space is $\mathbb{R}$
- A person (voter?) has an "ideal point" $p \in \mathbb{R}$
- Person prefers policies that are closer to his/her ideal point: receives utility from policy x

$$u(x) = -(x - p)^2$$
$$u'(x) = 2p - 2x$$

Algebra of Derivatives

Theorem

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ and both are differentiable at $x_0 \in \mathbb{R}$. Then,

i) $h(x) = f(x) + g(x)$ *is differentiable at* x_0 *and*

$$h^{'}(x_0) = f^{'}(x_0) + g^{'}(x_0)$$

ii) $h(x) = f(x)g(x)$ *is differentiable at* x_0 *and*

$$h^{'}(x_0) = f^{'}(x_0)g(x_0) + g^{'}(x_0)f(x_0)$$

iii) $h(x) = \frac{f(x)}{g(x)}$ *with* $g(x) \neq 0$ *then,*

$$h^{'}(x_0) = \frac{f^{'}(x_0)g(x_0) - g^{'}(x_0)f(x_0)}{g(x_0)^2}$$

Algebra of Derivatives: Examples

1) $f(x) = x^3 + 5x^2 + 4x$

$$\begin{aligned} f'(x) &= 3x^2 + 5 * 2x + 4 \\ &= 3x^2 + 10x + 4 \end{aligned}$$

2) $f(x) = sin(x)x^3$

$$\begin{aligned} f'(x) &= cos(X)x^3 + sin(x)3x^2 \\ &= cos(X)x^3 + 3sin(x)x^2 \end{aligned}$$

3) $f(x) = \frac{e^x}{x^3}$

$$\begin{aligned} f'(x) &= \frac{e^x x^3 - e^x 3x^2}{x^6} \\ &= \frac{e^x x^3 - 3e^x x^2}{x^6} \end{aligned}$$

Chain Rule

Common to have functions in functions

$$\begin{aligned} f(x) &= \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi}} \\ &= \frac{f(g(x))}{\sqrt{2\pi}} \end{aligned}$$

where $f(x) = e^{-g(x)}$, $g(x) = \frac{(x-\mu)^2}{2\sigma^2}$
To deal with this, we use the chain rule

Theorem

Suppose $g : \mathbb{R} \rightarrow \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$. Suppose both $f(x)$ and $g(x)$ are differentiable at x_0. Define $h(x) = g(f(x))$. Then,

$$h^{'}(x_0) = g^{'}(f(x_0))f^{'}(x_0)$$

Examples of Chain Rule

- $h(x) = e^{2x}$. $g(x) = e^{x}$. $f(x) = 2x$. So $h(x) = g(f(x)) = g(2x) = e^{2x}$. Taking derivatives, we have

$$h^{'}(x) = g^{'}(f(x))f^{'}(x) = e^{2x}2$$

- $h(x) = \log(\cos(x))$. $g(x) = log(x)$. $f(x) = \cos(x)$.
$h(x) = g(f(x)) = g(\cos(x)) = log(\cos(x))$

$$h^{'}(x) = g^{'}(f(x))f^{'}(x) = \frac{-1}{\cos(x)}\sin(x) = -\tan(x)$$

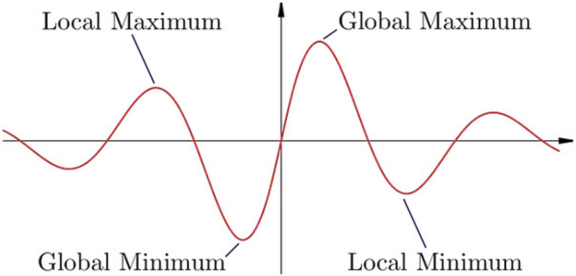
Derivative of a fraction

- We can use the chain rule to show that the product rule and the quotient rule are the same:

$$
\begin{aligned}
\frac{f(x)}{g(x)} &= f(x)(g(x))^{-1} \\
\frac{df(x)(g(x))^{-1}}{dx} &= f^{'}(x)(g(x))^{-1} + \frac{d(g(x))^{-1}}{dx} f(x) \\
&= f^{'}(x)(g(x))^{-1} + (-1)g(x)^{-2} g'(x) f(x) \\
&= \frac{f^{'}(x)}{g(x)} - \frac{g'(x)f(x)}{g(x)^2} \\
&= \frac{f^{'}(x)g(x) - g'(x)f(x)}{g(x)^2}
\end{aligned}
$$

- Derivative
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

Relative Maxima, Minima and Derivatives



- We use derivatives to find the maxima and minima.
- We need maxima (or minima) for statistical inference (what is **most** likely), modeling choices (what is **most preferred**), and for modeling systems (what is **stable**).
- The connection between maxima and derivatives is shown with the Mean Value Theorem (MVT).
- To get there we prove a special case of MVT called "Rolle's Theorem."

Aside: Naming Conventions of Theorems

- Stigler's Law of Eponomy.
  - "No scientific discovery is named for the person who discovered it."
- Rolle's Theorem was proved by Cauchy (1823), but first stated by a 12th century Indian mathematician Bhaskara II

Derivatives and local maxima/minima

Theorem

Rolle's Theorem Suppose $f : [a, b] \to \mathbb{R}$ and f is continuous on $[a, b]$ and differentiable on (a, b). Then if $f(a) = f(b) = 0$, there is $c \in (a, b)$ such that $f^{'}(c) = 0$.

- This claims that fixing two points at the same level, the function has to curve back at some point (it hits a max or min), where the tangent is flat.
- Proof intuition:
  - Consider a relative maximum c.
  - Find the left-hand (approaching from below) and right-hand limits (approaching from the above).
  - Show that the left hand limit can't be positive, and the right hand limit can't be negative.
  - Then all that is left is 0.

Formal Proof of Rolle's Theorem

Suppose $f : [a, b] \to \mathbb{R}$ and f is continuous on $[a, b]$ and differentiable on (a, b). If $f(x) = 0$ for all x, then $f'(x) = 0$ and we are done.
If not, there is some c that is either a min or a max.
If it is a max, then $f(x) - f(c) \leq 0$ By the definition of differentiability:

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} = f^{'}(c) \leq 0$$
$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} = f^{'}(c) \geq 0$$

In order for these two limits to be equal,

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} = \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

It must be that: $f^{'}(c) = 0$.

Mean Value Theorem

Theorem

If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b), then there is a $c \in (a,b)$ such that

$$f'(c) \quad = \quad \frac{f(b) - f(a)}{b - a}$$

- That is, if there is a line connecting two points of a function, the slope will be equal to the derivative somewhere between the two points.
- The proof strategy is to set up for an application of Rolle's theorem.

Proof of Mean Value Theorem

Consider $f(x)$ as your differentiable function. Construct a new function:

$$g(x) = f(x) - \frac{(f(b) - f(a))}{b - a}(x - a)$$

Note that: $g(a) = f(a)$

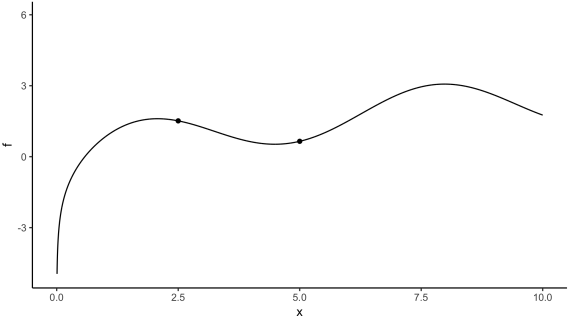
$$\begin{aligned} g(b) &= f(b) - \frac{(f(b) - f(a))(b - a)}{b - a} \\ &= f(a) \end{aligned}$$

So g(x) is like f(x), in that if you evaluate it at a we get $f(a)$, but if you evaluate it at b you still get $f(a)$. This means it starts and ends and the same place, so it must turn around.

Intuition for Proof Strategy

$$g(x) = f(x) - \frac{(f(b) - f(a))}{b - a}(x - a)$$

For instance:



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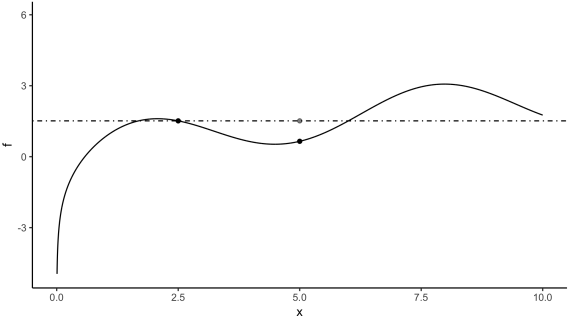

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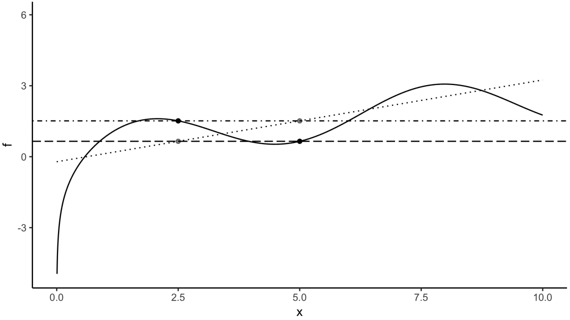

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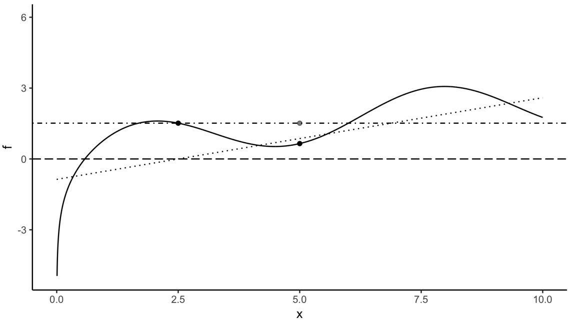

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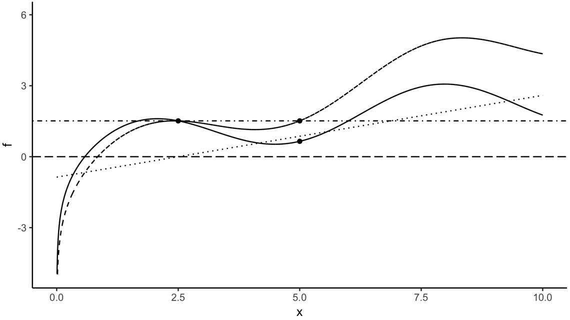


Applying Rolle's Theorem

$$g(x) = f(x) - \frac{(f(b) - f(a))}{b - a}(x - a)$$

$g'(x) = 0$ at some c by Rolle's Theorem

$$g'(x) = f'(x) - \frac{(f(b) - f(a))}{b - a}$$

$$g'(c) = 0 = f'(c) - \frac{(f(b) - f(a))}{b - a}$$

$$f'(c) = \frac{(f(b) - f(a))}{b - a}$$

Applications of Mean Value Theorem

- Speeding tickets: If the police can measure how long it takes me to get from point A to point B, at some point I was driving that speed.
- Used to derive popular approximations to functions (including Taylor expansion)
- We will use it in the proof of the Fundamental Theorem of Calculus relating derivatives to integrals.

- Derivative
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

Second Derivatives

If the derivative is a slope, the second derivative is the slope of a slope.

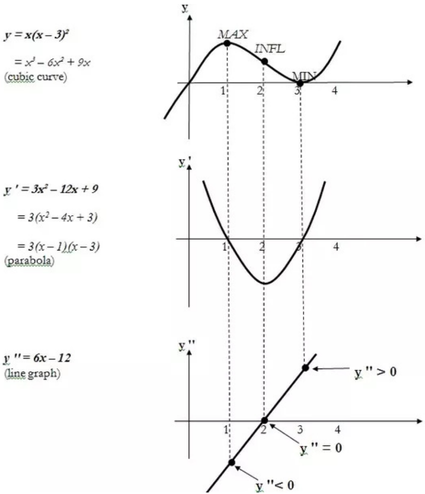
Definition

Suppose $f : \mathbb{R} \to \mathbb{R}$ is differentiable. Recall we write this as $f^{'}$ and suppose that $f^{'} : \mathbb{R} \to \mathbb{R}$. Then if the limit,

$$\lim_{x \to x_0} R(x) = \frac{f^{'}(x) - f^{'}(x_0)}{x - x_0}$$

exists, we call this the second derivative at x_0, $f^{''}(x_0)$.

Visualization of First and Second Derivative



Examples of Second Derivatives

$$
\begin{aligned}
f(x) &= x \\
f^{'}(x) &= 1 \\
f^{''}(x) &= 0
\end{aligned}
$$

$$
\begin{aligned}
f(x) &= e^x \\
f^{'}(x) &= e^x \\
f^{''}(x) &= e^x
\end{aligned}
$$

$$
\begin{aligned}
f(x) &= \log(x) \\
f^{'}(x) &= \frac{1}{x} = x^{-1} \\
f^{''}(x) &= -\frac{1}{x^2}
\end{aligned}
$$

Second Derivatives

If the derivative is a slope, the second derivative is the slope of a slope.

Definition

Suppose $f : \mathbb{R} \to \mathbb{R}$ *is differentiable. Recall we write this as* $f^{'}$ *and suppose that* $f^{'} : \mathbb{R} \to \mathbb{R}$*. Then if the limit,*

$$\lim_{x \to x_0} R(x) = \frac{f^{'}(x) - f^{'}(x_0)}{x - x_0}$$

exists, we call this the second derivative at x_0, $f^{''}(x_0)$.

Going forward

- Your job is to memorize the product, quotient and chain rules of derivatives.
- We will see how the second derivative is used to evaluate points as relative minima and maxima.